Week 8: Database Relationship Analysis using Joins

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Scenario:

The e-commerce company maintains customer, product, order, and payment information in its database. In this week, SQL joins are implemented to combine related data from multiple tables using INNER JOIN, LEFT JOIN, and RIGHT JOIN. Queries are used to identify customers without purchases, products without sales, and orders with their payment details. Customer purchase history and multi-table business reports are also generated to analyse customer spending, order counts, sales, and payment information.

1. INNER JOIN

INNER JOIN returns only matching records from both tables.

1. Display customer details along with their orders

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS
  3  FROM Customer C
  4  INNER JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  ORDER BY O.ORDER_ID;

CUSTOMER_I CUSTOMER_NAME                                     ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
C001       Rahul                                                 101
15-SEP-26          499 Pending

C002       Ananya                                                102
16-SEP-26          699 Shipped

C003       Vikram                                                103
17-SEP-26          299 Delivered


CUSTOMER_I CUSTOMER_NAME                                     ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
C004       Meena                                                 104
18-SEP-26         1198 Pending

C005       Karthik                                               105
19-SEP-26          799 Shipped

C002       Ananya                                                107
21-SEP-26          599 Pending

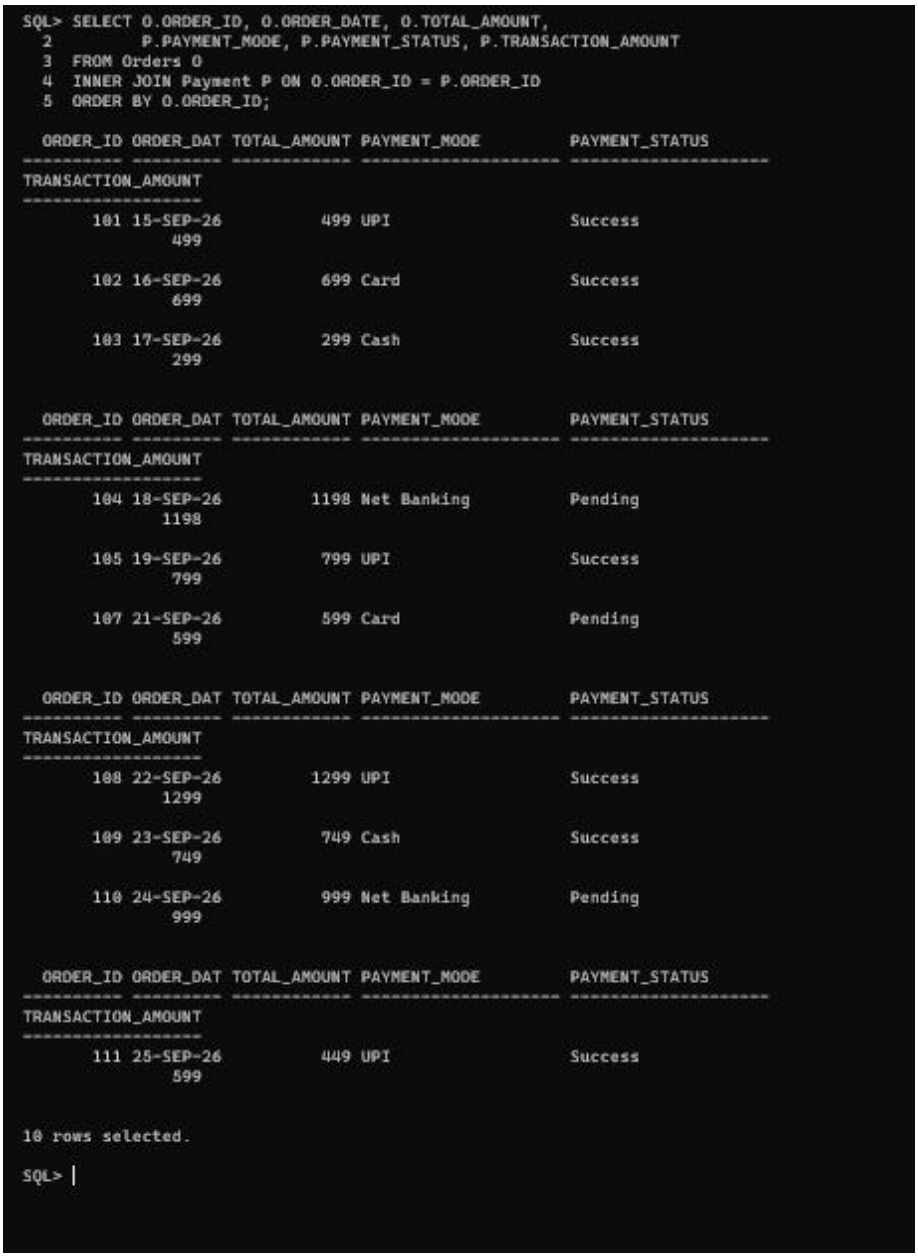

CUSTOMER_I CUSTOMER_NAME                                     ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
C003       Vikram                                                108
22-SEP-26         1299 Shipped

C004       Meena                                                 109
23-SEP-26          749 Delivered

C005       Karthik                                               110
24-SEP-26          999 Pending


CUSTOMER_I CUSTOMER_NAME                                     ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
C001       Rahul                                                 111
25-SEP-26          449 Shipped


10 rows selected.
```



```
SQL> SELECT O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT,
  2         P.PAYMENT_MODE, P.PAYMENT_STATUS, P.TRANSACTION_AMOUNT
  3  FROM Orders O
  4  INNER JOIN Payment P ON O.ORDER_ID = P.ORDER_ID
  5  ORDER BY O.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
       101 15-SEP-26          499 UPI                  Success
               499

       102 16-SEP-26          699 Card                 Success
               699

       103 17-SEP-26          299 Cash                 Success
               299


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
       104 18-SEP-26         1198 Net Banking          Pending
              1198

       105 19-SEP-26          799 UPI                  Success
               799

       107 21-SEP-26          599 Card                 Pending
               599


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
       108 22-SEP-26         1299 UPI                  Success
              1299

       109 23-SEP-26          749 Cash                 Success
               749

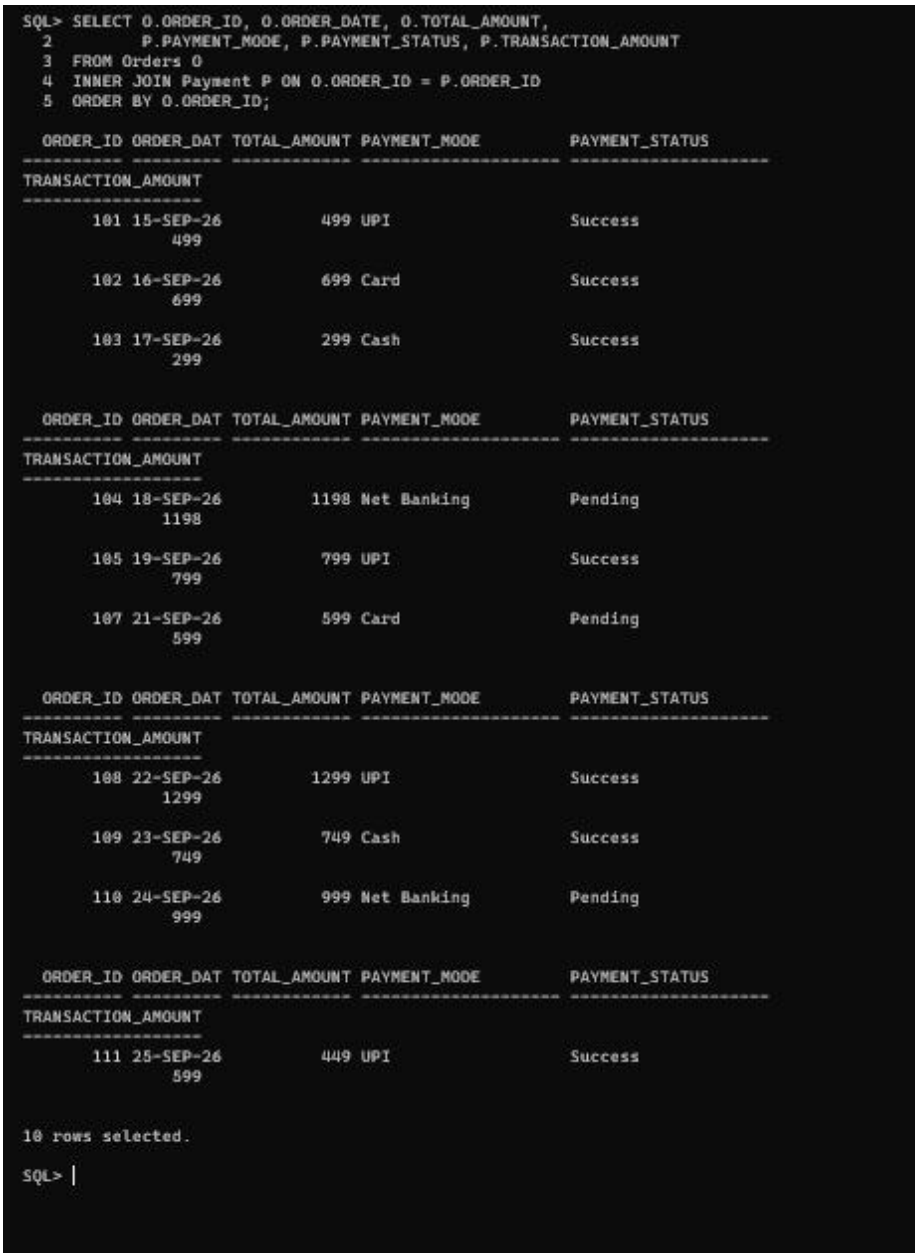
       110 24-SEP-26          999 Net Banking          Pending
               999


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
       111 25-SEP-26          449 UPI                  Success
               599


10 rows selected.

SQL>
```

2. Display order details with payment information



```
SQL> SELECT O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT,
  2         P.PAYMENT_MODE, P.PAYMENT_STATUS, P.TRANSACTION_AMOUNT
  3  FROM Orders O
  4  INNER JOIN Payment P ON O.ORDER_ID = P.ORDER_ID
  5  ORDER BY O.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
       101 15-SEP-26          499 UPI                  Success
               499

       102 16-SEP-26          699 Card                 Success
               699

       103 17-SEP-26          299 Cash                 Success
               299


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
       104 18-SEP-26         1198 Net Banking          Pending
              1198

       105 19-SEP-26          799 UPI                  Success
               799

       107 21-SEP-26          599 Card                 Pending
               599


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
       108 22-SEP-26         1299 UPI                  Success
              1299

       109 23-SEP-26          749 Cash                 Success
               749

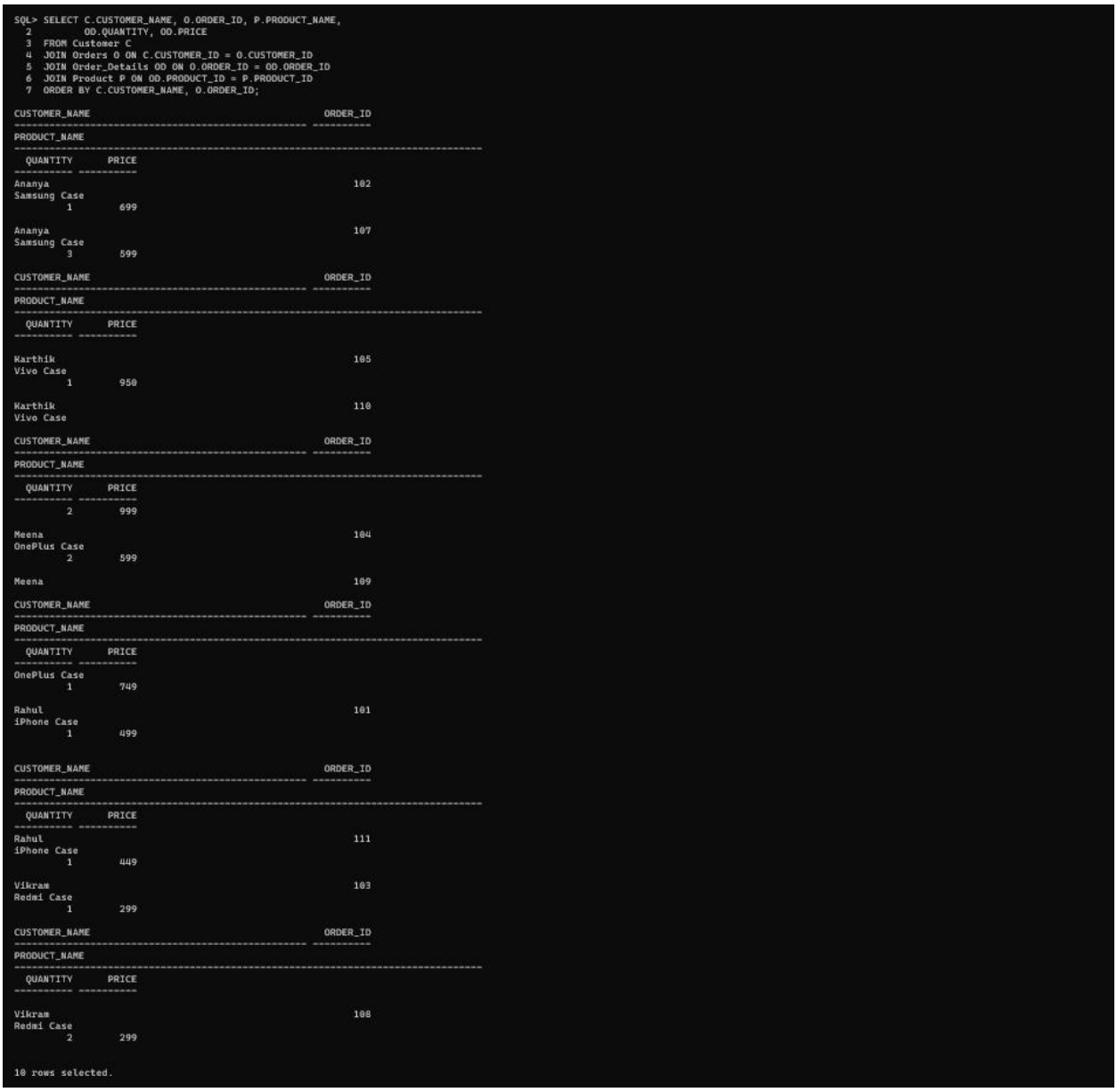
       110 24-SEP-26          999 Net Banking          Pending
               999


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
       111 25-SEP-26          449 UPI                  Success
               599


10 rows selected.

SQL>
```

3. Retrieve products purchased by customers



```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID, P.PRODUCT_NAME,
  2         OD.QUANTITY, OD.PRICE
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_ID
  6  JOIN Product P ON OD.PRODUCT_ID = P.PRODUCT_ID
  7  ORDER BY C.CUSTOMER_NAME, O.ORDER_ID;

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Ananya                                                102
Samsung Case
         1        699

Ananya                                                107
Samsung Case
         3        599

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Karthik                                               105
Vivo Case
         1        950

Karthik                                               110
Vivo Case

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         2        999

Meena                                                 104
OnePlus Case
         2        599

Meena                                                 109

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
OnePlus Case
         1        749

Rahul                                                 101
iPhone Case
         1        499


CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Rahul                                                 111
iPhone Case
         1        449

Vikram                                                103
Redmi Case
         1        299

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Vikram                                                108
Redmi Case
         2        299


10 rows selected.
```

2. LEFT JOIN

4. Display all customers, including those without orders

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT
  3  FROM Customer C
  4  LEFT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  ORDER BY C.CUSTOMER_ID;

CUSTOMER_I CUSTOMER_NAME                                      ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
C001       Rahul                                                 101
15-SEP-26          499

C001       Rahul                                                 111
25-SEP-26          449

C002       Ananya                                                102
16-SEP-26          699


CUSTOMER_I CUSTOMER_NAME                                      ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
C002       Ananya                                                107
21-SEP-26          599

C003       Vikram                                                103
17-SEP-26          299

C003       Vikram                                                108
22-SEP-26         1299


CUSTOMER_I CUSTOMER_NAME                                      ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
C004       Meena                                                 104
18-SEP-26         1198

C004       Meena                                                 109
23-SEP-26          749

C005       Karthik                                               105
19-SEP-26          799


CUSTOMER_I CUSTOMER_NAME                                      ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
C005       Karthik                                               110
24-SEP-26          999


10 rows selected.

SQL>
```

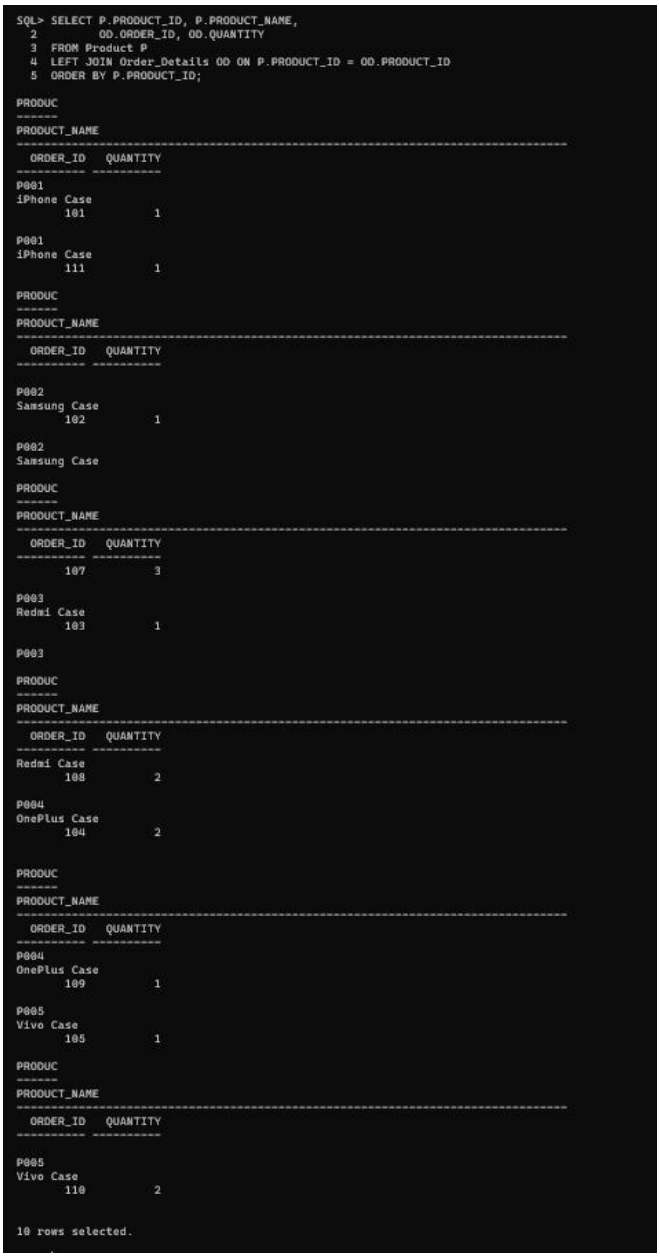
5. Find customers without any purchases

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME
  2  FROM Customer C
  3  LEFT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  4  WHERE O.ORDER_ID IS NULL;

CUSTOMER_I CUSTOMER_NAME
---------- --------------------------------------------------
C006       Priya

SQL>
```

6. Display all products, including products with no sales



```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME,
  2         OD.ORDER_ID, OD.QUANTITY
  3  FROM Product P
  4  LEFT JOIN Order_Details OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  5  ORDER BY P.PRODUCT_ID;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
P001
iPhone Case
       101          1

P001
iPhone Case
       111          1

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------

P002
Samsung Case
       102          1

P002
Samsung Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
       107          3

P003
Redmi Case
       103          1

P003

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
Redmi Case
       108          2

P004
OnePlus Case
       104          2


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
P004
OnePlus Case
       109          1

P005
Vivo Case
       105          1

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------

P005
Vivo Case
       110          2


10 rows selected.
```

3. RIGHT JOIN

7. Display all orders with customer information

```
Connected.
SQL> SELECT O.ORDER_ID,
  2         O.ORDER_DATE,
  3         O.TOTAL_AMOUNT,
  4         C.CUSTOMER_ID,
  5         C.CUSTOMER_NAME
  6  FROM CUSTOMER C
  7  RIGHT JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  ORDER BY O.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTOMER_I
---------- --------- ------------ ----------
CUSTOMER_NAME
--------------------------------------------------
       101 15-SEP-26          499 C001
Rahul

       102 16-SEP-26          699 C002
Ananya

       103 17-SEP-26          299 C003
Vikram


  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTOMER_I
---------- --------- ------------ ----------
CUSTOMER_NAME
--------------------------------------------------
       104 18-SEP-26         1198 C004
Meena

       105 19-SEP-26          799 C005
Karthik

       107 21-SEP-26          599 C002
Ananya


  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTOMER_I
---------- --------- ------------ ----------
CUSTOMER_NAME
--------------------------------------------------
       108 22-SEP-26         1299 C003
Vikram

       109 23-SEP-26          749 C004
Meena

       110 24-SEP-26          999 C005
Karthik


  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTOMER_I
---------- --------- ------------ ----------
CUSTOMER_NAME
--------------------------------------------------
       111 25-SEP-26          449 C001
Rahul


10 rows selected.

SQL>
```

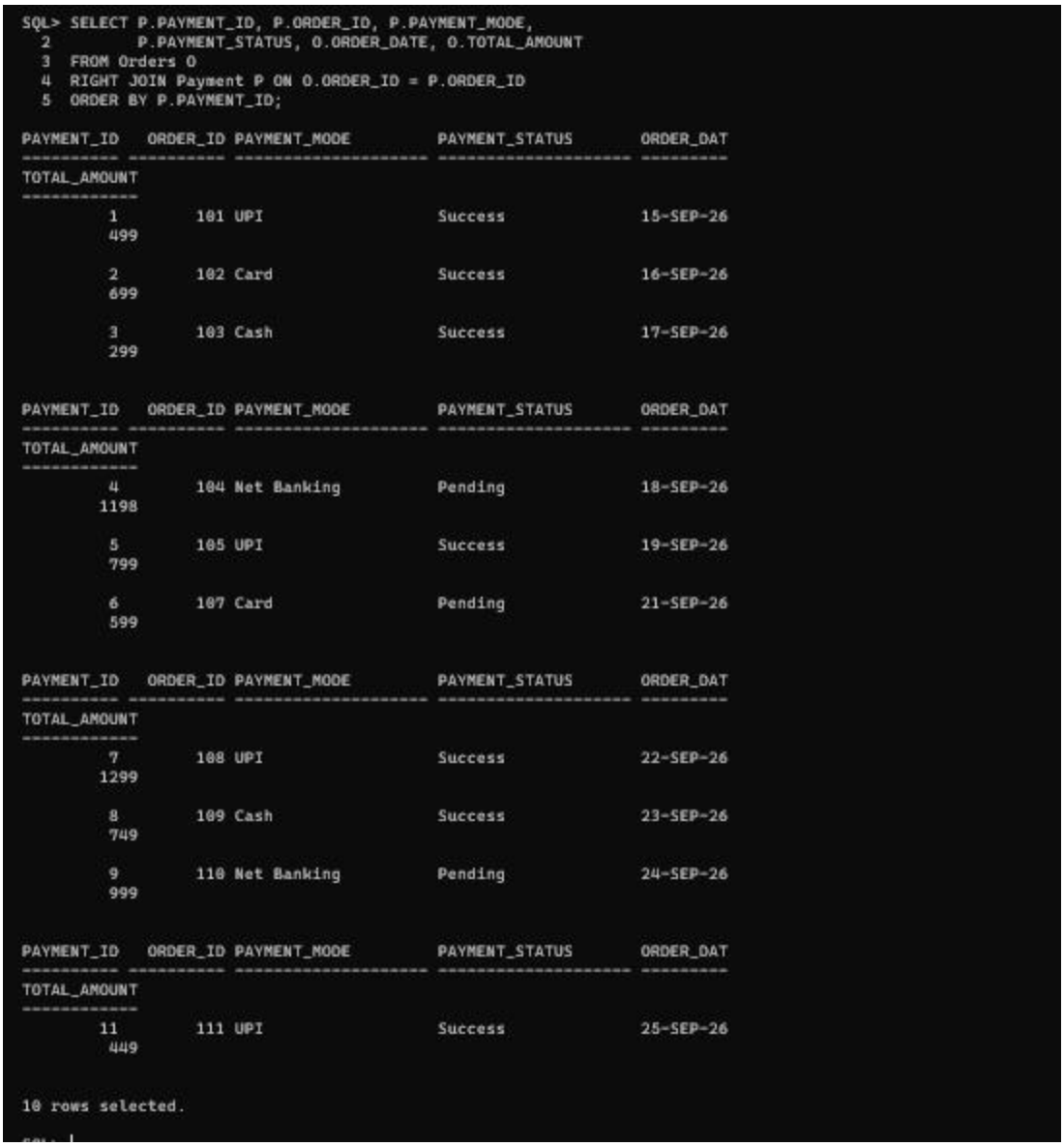
8. Find orders where customer details are missing

```
SQL> SELECT O.ORDER_ID, O.CUSTOMER_ID, O.ORDER_DATE
  2  FROM Customer C
  3  RIGHT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  4  WHERE C.CUSTOMER_ID IS NULL;

no rows selected

SQL>
```

9. Display all payment records with order details



```
SQL> SELECT P.PAYMENT_ID, P.ORDER_ID, P.PAYMENT_MODE,
  2         P.PAYMENT_STATUS, O.ORDER_DATE, O.TOTAL_AMOUNT
  3  FROM Orders O
  4  RIGHT JOIN Payment P ON O.ORDER_ID = P.ORDER_ID
  5  ORDER BY P.PAYMENT_ID;

PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
         1        101 UPI                  Success              15-SEP-26
       499

         2        102 Card                 Success              16-SEP-26
       699

         3        103 Cash                 Success              17-SEP-26
       299


PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
         4        104 Net Banking          Pending              18-SEP-26
      1198

         5        105 UPI                  Success              19-SEP-26
       799

         6        107 Card                 Pending              21-SEP-26
       599


PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
         7        108 UPI                  Success              22-SEP-26
      1299

         8        109 Cash                 Success              23-SEP-26
       749

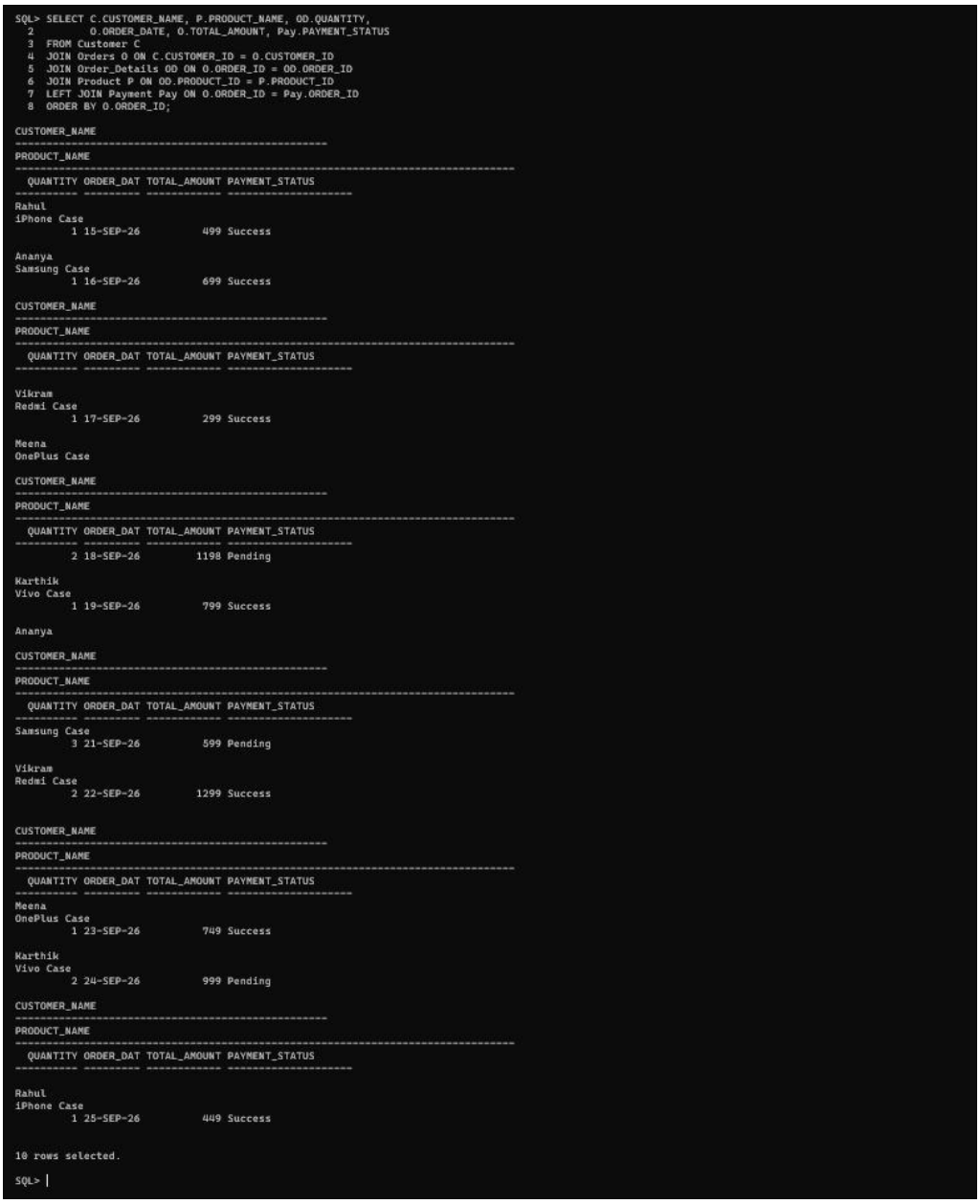
         9        110 Net Banking          Pending              24-SEP-26
       999


PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
        11        111 UPI                  Success              25-SEP-26
       449


10 rows selected.
```

4. Retrieve Complete Order Details

10. Generate a complete order report



```
SQL> SELECT C.CUSTOMER_NAME, P.PRODUCT_NAME, OD.QUANTITY,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT, Pay.PAYMENT_STATUS
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_ID
  6  JOIN Product P ON OD.PRODUCT_ID = P.PRODUCT_ID
  7  LEFT JOIN Payment Pay ON O.ORDER_ID = Pay.ORDER_ID
  8  ORDER BY O.ORDER_ID;

CUSTOMER_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Rahul
iPhone Case
         1 15-SEP-26          499 Success

Ananya
Samsung Case
         1 16-SEP-26          699 Success

CUSTOMER_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------

Vikram
Redmi Case
         1 17-SEP-26          299 Success

Meena
OnePlus Case

CUSTOMER_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
         2 18-SEP-26         1198 Pending

Karthik
Vivo Case
         1 19-SEP-26          799 Success

Ananya

CUSTOMER_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Samsung Case
         3 21-SEP-26          599 Pending

Vikram
Redmi Case
         2 22-SEP-26         1299 Success


CUSTOMER_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Meena
OnePlus Case
         1 23-SEP-26          749 Success

Karthik
Vivo Case
         2 24-SEP-26          999 Pending

CUSTOMER_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------

Rahul
iPhone Case
         1 25-SEP-26          449 Success


10 rows selected.

SQL>
```

5. Display Customer Purchase History

11. Display all products purchased by a customer

```
SQL> SELECT C.CUSTOMER_NAME, P.PRODUCT_NAME,
  2         O.ORDER_ID, OD.QUANTITY, OD.PRICE
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_ID
  6  JOIN Product P ON OD.PRODUCT_ID = P.PRODUCT_ID
  7  WHERE C.CUSTOMER_ID = 'C001'
  8  ORDER BY O.ORDER_DATE DESC;

CUSTOMER_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY      PRICE
---------- ---------- ----------
Rahul
iPhone Case
       111          1        449

Rahul
iPhone Case
       101          1        499

CUSTOMER_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY      PRICE
---------- ---------- ----------


SQL>
```

12. Calculate total amount spent by each customer

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
  2         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  3  FROM Customer C
  4  LEFT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY NUMBER_OF_ORDERS DESC;

CUSTOMER_I CUSTOMER_NAME                                      NUMBER_OF_ORDERS
---------- -------------------------------------------------- ----------------
C001       Rahul                                                             2
C002       Ananya                                                            2
C004       Meena                                                             2
C003       Vikram                                                            2
C005       Karthik                                                           2
C006       Priya                                                             0

6 rows selected.

SQL>
```

13. Count the number of orders placed by each customer

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
  2         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  3  FROM Customer C
  4  LEFT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY NUMBER_OF_ORDERS DESC;

CUSTOMER_I CUSTOMER_NAME                                      NUMBER_OF_ORDERS
---------- -------------------------------------------------- ----------------
C001       Rahul                                                             2
C002       Ananya                                                            2
C004       Meena                                                             2
C003       Vikram                                                            2
C005       Karthik                                                           2
C006       Priya                                                             0

6 rows selected.

SQL>
```

14. Display the latest purchase details of each customer

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  WHERE O.ORDER_DATE = (
  6      SELECT MAX(O2.ORDER_DATE)
  7      FROM Orders O2
  8      WHERE O2.CUSTOMER_ID = C.CUSTOMER_ID
  9  )
 10  ORDER BY C.CUSTOMER_NAME;

CUSTOMER_NAME                                     ORDER_ID ORDER_DAT
------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
Ananya                                               107 21-SEP-26
         599 Pending

Karthik                                              110 24-SEP-26
         999 Pending

Meena                                                109 23-SEP-26
         749 Delivered


CUSTOMER_NAME                                     ORDER_ID ORDER_DAT
------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
Rahul                                                111 25-SEP-26
         449 Shipped

Vikram                                               108 22-SEP-26
        1299 Shipped


SQL>
```

6. Multi-Table Business Reports Report 1: Customer Order Report

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.ORDER_STATUS
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  ORDER BY O.ORDER_ID;

CUSTOMER_NAME                                     ORDER_ID ORDER_DAT
------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Rahul                                                  101 15-SEP-26
Pending

Ananya                                                 102 16-SEP-26
Shipped

Vikram                                                 103 17-SEP-26
Delivered


CUSTOMER_NAME                                     ORDER_ID ORDER_DAT
------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Meena                                                  104 18-SEP-26
Pending

Karthik                                                105 19-SEP-26
Shipped

Ananya                                                 107 21-SEP-26
Pending


CUSTOMER_NAME                                     ORDER_ID ORDER_DAT
------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Vikram                                                 108 22-SEP-26
Shipped

Meena                                                  109 23-SEP-26
Delivered

Karthik                                                110 24-SEP-26
Pending


CUSTOMER_NAME                                     ORDER_ID ORDER_DAT
------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Rahul                                                  111 25-SEP-26
Shipped


10 rows selected.
```

Report 2: Sales Report

```
SQL> SELECT P.PRODUCT_NAME,
  2         SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM Product P
  5  JOIN Order_Details OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  7  ORDER BY TOTAL_REVENUE DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Vivo Case
            3          2948

Samsung Case
            4          2496

OnePlus Case
            3          1947


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
iPhone Case
            2           948

Redmi Case
            3           897


SQL>
```

Report 3: Payment Analysis Report

```
SQL> SELECT PAYMENT_MODE,
  2         COUNT(*) AS NUMBER_OF_TRANSACTIONS,
  3         SUM(CASE WHEN PAYMENT_STATUS = 'Completed'
  4                  THEN 1 ELSE 0 END) AS SUCCESSFUL_PAYMENTS
  5  FROM Payment
  6  GROUP BY PAYMENT_MODE
  7  ORDER BY PAYMENT_MODE;

PAYMENT_MODE         NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ---------------------- --------------------
Card                                      2                    0
Cash                                      2                    0
Net Banking                               2                    0
UPI                                       4                    0

SQL>
```

Report 4: Customer Purchase Analysis

Top purchasing customers

```
SQL> SELECT C.CUSTOMER_NAME,
  2         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME                                      TOTAL_SPENDING
-------------------------------------------------- --------------
Meena                                                        1947
Karthik                                                      1798
Vikram                                                       1598
Ananya                                                       1298
Rahul                                                         948

SQL>
```

• Customers with maximum orders

```
SQL> SELECT C.CUSTOMER_NAME,
  2         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME                                      TOTAL_SPENDING
-------------------------------------------------- --------------
Meena                                                        1947
Karthik                                                      1798
Vikram                                                       1598
Ananya                                                       1298
Rahul                                                         948

SQL>
```

• Customers with maximum orders

```
SQL> SELECT C.CUSTOMER_NAME,
  2         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY NUMBER_OF_ORDERS DESC;

CUSTOMER_NAME                                      NUMBER_OF_ORDERS
-------------------------------------------------- ----------------
Ananya                                                            2
Rahul                                                             2
Meena                                                             2
Karthik                                                           2
Vikram                                                            2

SQL>
```

Customers with highest spending

```
SQL> SELECT C.CUSTOMER_NAME,
  2         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  HAVING SUM(O.TOTAL_AMOUNT) = (
  7      SELECT MAX(TOTAL_SPENT)
  8      FROM (
  9          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
 10          FROM Orders
 11          GROUP BY CUSTOMER_ID
 12      )
 13  );

CUSTOMER_NAME                                      TOTAL_SPENDING
-------------------------------------------------- --------------
Meena                                                        1947

SQL>
```

